



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Probability Distributions (Theory)
Course Code: STAT-316

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2 Attempt the following questions.

- Find the mean deviation of the Exponential distribution. (3)
- Describe the relationship between the Gamma distribution and Exponential distribution. (2)
- Find the mean and variance of the rectangular distribution on $(-2, 2)$ (3)
- Show that $t^2_{(n)} = F_{(1,n)}$ (3)
- Find moment generating function of the Student-t distribution. (3)
- If $X \sim N(\mu, \sigma^2)$ then find the mean and variance of $Y = aX + b$. (3)
- Define distribution function and its properties. (3)

Q.3.a) If "X" follows two parameter Gamma distribution with density function

$$f(x) = \frac{x^{\alpha-1} e^{-\frac{x}{\beta}}}{\Gamma(\alpha) \beta^{\alpha}} ; 0 < x < \infty, \alpha > 0, \beta > 0.$$

Derive its moment generating function and use it to find first four cumulants and γ_1 and γ_2

- Show that marginal density of "X" in Bivariate Normal distribution follows univariate Normal distribution. (7+3)

Q.4 State and derive the probability density function of t-distribution.

(10)

Q.5.a) If X has an Exponential distribution given by

$$f(x) = \frac{1}{2} e^{-\frac{x}{2}} ; 0 \leq x < \infty.$$

What are the mean, variance and m.g.f. of X? Also calculate

$$P(X < 3) \text{ and } P(X > 5 | X > 2).$$

- Five observations x_1, x_2, x_3, x_4, x_5 are drawn at random from the distribution

$$f(x) = 5e^{-5x} ; 0 \leq x < \infty.$$

What is the distribution of (i) The smallest observation (ii) The largest observation?

(7+3)



UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Probability Distributions (Theory)

TIME ALLOWED: 2 Hrs. & 45 Mins.

Course Code: STAT-316 Part – II

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2. a) Find the mean and variance of Pareto distribution. (5)

b) If $X_1, X_2, \dots, X_n$ are 'n' independent r.v's having exponential distribution having same parameter ' θ ' i.e. $f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$ (5)

c) Find the pdf of $Y = X_1 + X_2 + \dots + X_n$

If f is a F distributed random variable with n_1 & n_2 d.f then show that (5)

$y = \left(1 + \frac{n_1}{n_2} F\right)^{-\frac{n_2}{2}}$ has beta distribution of first kind.

d) If X is a random variable with p.d.f as (5)

$$f(x) = \frac{1}{2} \quad -1 \leq x \leq +1$$

Find the p.d.f of $Y = X^2$

Q.3. Derive χ^2 - distribution. (10)

Q.4. Find the coefficient of skewness and kurtosis of lognormal distribution. (10)

Q.5. Let X_1 and X_2 be two independent standard normal variables. Then obtain the distribution of $Z = \frac{X_1}{X_2}$. (10)

**ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED**

- Q2 i) Show that the square of t- variate with n degree of freedom is (20)
distributed as F with 1 and n degree of freedom.
- ii) Let $Y_1 < Y_2 < Y_3 < Y_4$ denotes the order statistic of a random sample of size 4 from distribution having p.d.f.
$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution of Y_3 .
- iii) Let X and Y are independent random variables with joint p.d.f.
$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E(XY)$ and $E(X + Y)$.
- iv) If the random variable X has normal distribution with mean 'm' and variance ' σ^2 '. Obtain the distribution of $Y = aX + b$.
- v) Find mean and variance of Pareto Distribution.
- Q3. Define lognormal distribution and obtain its rth moment about origin. (8)
If $\log_e x$ is normally disturbed with $\mu = 1, \sigma^2 = 4$. Find $P(\frac{1}{2} < X < 2)$.
- Q4. Let X_1 and X_2 be two independent standard normal variables. Let (7)
 $Y_1 = X_1 + X_2$ and $Y_2 = \frac{X_1}{X_2}$. Find the marginal p.d.f. of Y_2 .
- Q5. Derive F-distribution. (8)
- Q6. Derive characteristic function of Cauchy distribution. (7)



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2 Write short answer on the following questions.

(4 × 5 = 20)

(i) Find the distribution of $Y = X^2$, given that

$$f(x) = \frac{1}{\pi(1+x^2)} \quad -\infty \leq x \leq \infty$$

(ii) Find mean of log normal distribution

(iii) Assume that $X_1, X_2, \dots, X_n$ are independent r.v's and $X_i \sim N(\mu_i, \sigma_i^2)$. Find the p.d.f of $Y = \sum a_i X_i$, where a_i 's are constants.

(iv) Prove Chebyshev's Inequality.

Q.3. Find γ_1 and γ_2 of Gamma distribution.

(10)

where

$$f(x) = \frac{1}{\beta^{\alpha+1} \Gamma_{\alpha+1}} x^{\alpha} e^{-x/\beta} \quad x \geq 0$$

0 elsewhere

Q.4. Derive F-distribution.

(10)

Q.5. Let X_1 and X_2 be two independent standard normal variables. Let $Y_1 = X_1 + X_2$ and

$Y_2 = \frac{X_1}{X_2}$. Find the marginal p.d.f. of Y_2 .

(10)



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program / Sixth Semester – 2021

Paper: Probability Distributions (Theory)
Course Code: STAT-316 Part – II

Roll No.

Time: 2 Hrs. 45 Min. Marks: 60

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2 Attempt the following questions.

- Find the mean deviation of the normal distribution. (3)
- Describe the relationship between the Cauchy distribution and Student t-distribution. (2)
- Find the expression for the r^{th} moments of Weibull distribution. (3)
- Find the mean and variance of the Beta distribution. (3)
- Find moment generating function of the Chi-square distribution. (3)
- Find the median of the Cauchy distribution. (3)
- Show that Chi-square distribution approaches to standard normal distribution when degree of freedom approaches to infinity. (3)

- Q.3.a) If "X" follows exponential distribution then derive its moment generating function and use it to find first four cumulants, skewness and kurtosis.
- b) Show that marginal density of X in Bivariate Normal distribution follows univariate Normal distribution. (7+3)

Q.4 State and derive the probability density function of F-distribution. (10)

Q.5.a) Let x and y be independent gamma variates with parameters (α_1, β_1) and (α_2, β_2) then obtain the distribution of $z = x + y$.

- b) Five observations x_1, x_2, x_3, x_4, x_5 are drawn at random from the distribution

$$f(x) = 20e^{-20x} ; 0 \leq x \leq \infty.$$

Find the distributions of the following (i) the smallest observation (ii) the largest observation? (7+3)

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

(i) If X is normally distributed with mean μ and variance σ^2 then show that

$$E|X - \mu| = \sqrt{\frac{2}{\pi}} \sigma$$

(ii) The two continuous random variables are jointly distributed as

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4} & \text{for } 0 < x < 2; \quad 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the conditional probability density functions $f(x|y)$ and $f(y|x)$. Are X and Y statistically independent?

(iii) Let X is normally distributed with mean zero and variance, σ^2 . Find the probability distribution of $Y = |X|$

(iv) Show that t -distribution approaches to standard normal distribution as degrees of freedom tends to infinity.

(v) If random variable F has F -distribution with v_1 and v_2 degrees of freedom then show that $y = \frac{v_1}{v_2} F$ has a Beta distribution of second kind.

(vi) Let $Y_1 < Y_2 < Y_3 < Y_4$ denote the order statistics of a random sample of size 4 from the distribution having the probability density function

$$f(x) = \lambda e^{-\lambda x} \text{ for } 0 < x < \infty$$

Find the distribution of the smallest observation.

Answer the following questions.

(3x10=30)

Q.2 Let the random variables X and Y have a bivariate normal distribution with parameters $m_1, m_2, \sigma_1^2, \sigma_2^2$, and ρ , then show that the conditional distribution of X given $Y = y$ is normal with mean $[m_1 + \rho \frac{\sigma_1}{\sigma_2} (y - m_2)]$ and variance $\sigma_1^2 (1 - \rho^2)$.

Q.3 If X and Y are two independent gamma variates with parameters $(\alpha, 1)$ and $(\beta, 1)$, then find the distribution of $Z = X + Y$

Q.4 State and derive the F -distribution.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- i) Define moment generating function of a random variable? Describe the significance of moments in a probability distribution.
- ii) Derive the variance of the normal distribution.
- iii) Derive r^{th} moment about origin, μ'_r , of the Pareto distribution and hence find its mean and variance.
- iv) Show that the area under the curve of lognormal distribution is unity.
- v) Let the random variable X is normally distributed with mean zero and variance one, find probability distribution of $Y = X^2$
- vi) Derive moment generating function of chi-square distribution and hence find its mean and variance.

Answer the following Questions.

(3x10=30)

- Q.2 (a) Derive moment generating function of the normal distribution. (7)
- (b) Find the distribution function of exponential distribution (3)
- Q.3 Derive the probability density function of t -distribution (10)
- Q.4 Let $Y_1 < Y_2 < Y_3 < Y_4 < Y_5$ denotes the order statistics of a random sample of size 5 from the distribution having the probability density function

$$f(x) = e^{-x} \text{ for } x > 0$$

Show that the statistics $Z_1 = Y_2$ and $Z_2 = Y_4 - Y_2$ are statistically independent (10)



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- a) Find moment generating function of the Chi-square distribution.
- b) Show that for the rectangular distribution,

$$f(x) = \frac{1}{2a} \quad -a < x < a$$

The m.g.f. about origin is $\frac{1}{at} \sinh at$. Also show that the moments of even order given by

$$\mu_{2n} = \frac{a^{2n}}{(2n+1)}.$$

- c) Derive the mean deviation of the exponential distribution.
- d) If X has standard normal distribution. Find the distribution of $Y = X^2$.
- e) Derive the distribution of r th (Y_r) and s th (Y_s) order statistic.
- f) Assume that $X_1, X_2, \dots, X_n$ are independent random variables and $X_i \sim N(\mu_i, \sigma_i^2)$. Find the p.d.f. of $Y = \sum a_i X_i$ where a_i 's are constants

Answer the following questions.

(3x10=30)

Q2. Derive the mean and variance of t-distribution.

Q3. Let $Y_1 < Y_2 < Y_3 < Y_4$ denotes the order statistic of a random sample of size 4 from distribution having p.d.f.

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution of Y_3 and compute $P(\frac{1}{2} < Y_3)$.

Q4. Write a note on Lognormal distribution. Also, find the mean and variance of lognormal distribution.